

Dental Cements

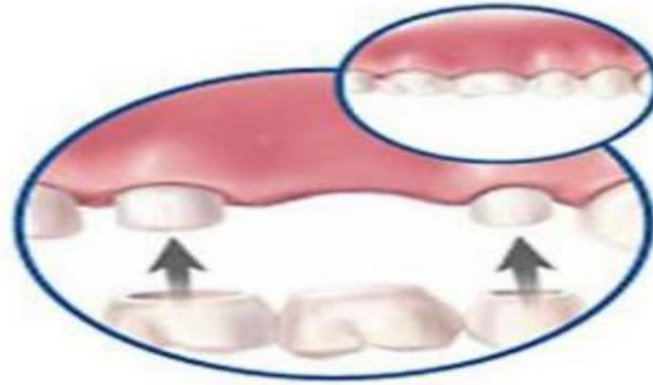
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Application of dental cements

1. Cavity **lining and base** under restorations.
2. **Luting** applications i.e. cementation of crowns, bridges, inlays, onlays and orthodontic brackets.
“temporary or permanent”.
3. Some cements are specifically formulated to be used as **temporary or permanent filling** materials.
4. Certain other cements for specialized purposes in the **endodontic, periodontic and surgical** fields of dentistry.

Application of dental cements



Ideal Requirements of dental Cements

1. Biological properties:

- a) **Non-toxic** and **non-irritant** to the pulp and other oral tissues.
- b) It should be **obtundent** (alleviate pulpal irritation & pain), **palliative**, and **sedative** to oral tissues.
- c) It should be **bacteriostatic**.
- d) It should provide **thermal insulation**.
- e) It should provide **chemical insulation**.
- f) It should provide **electrical insulation**.

Ideal Requirements of dental Cements

2. Rheological properties:

- a) The luting cement should have thin film thickness.
- b) The cement should have a good working time and short setting time.

Ideal Requirements of dental Cements

2. Rheological properties:

The cement film thickness depends on:

1. **The powder particle size:** finer powder particle size → a lower film thickness.
2. **The viscosity of the mix:** depends on the **P/L ratio**.
Using more liquid for mixing results in a less viscous (more flowable) cement mix → easily flushed into a low film thickness during seating of the restoration.
3. **The amount of pressure applied** on the restoration during cementation: Using more pressure allows better seating with less film thickness.

Ideal Requirements of dental Cements

3. Solubility:

- ▶ The cement should be **insoluble** in oral fluids to prevent leakage, loosening of the restoration and recurrent caries.

4. Mechanical properties:

- ▶ Mechanical requirements vary according to uses of dental cements.

e.g. if used as a liner should develop strength rapidly **Why??**
to enable filling material to be packed over it.

- ▶ The strength of the cement depends on its composition and W/P %.
- ▶ Base under amalgam should have sufficient strength **why??**
To withstand condensation force of amalgam.

Ideal Requirements of dental Cements

5. Bonding:

- ▶ The cement should bond to tooth structure and to restorations by either:
 - a) **Mechanical interlocking:** the freshly mixed cement flows into the microscopic irregularities on the tooth and restoration surface then hardens.
 - b) **Chemical bonding:** the freshly mixed cement forms a chemical bond with the tooth and the restoration surface.
- ▶ Chemical adhesion is considered as **gold standard** for bonding.
- ▶ Cements with chemical bonding are described as “**adhesive**”.

Ideal Requirements of dental Cements

5. Optical properties:

The optical requirements also depend on the application of the dental cement:

1. If used for cementing opaque restoration e.g. all metallic crown: either opaque or translucent cement.
2. If used for cementing translucent restoration: only translucent cement can be used.

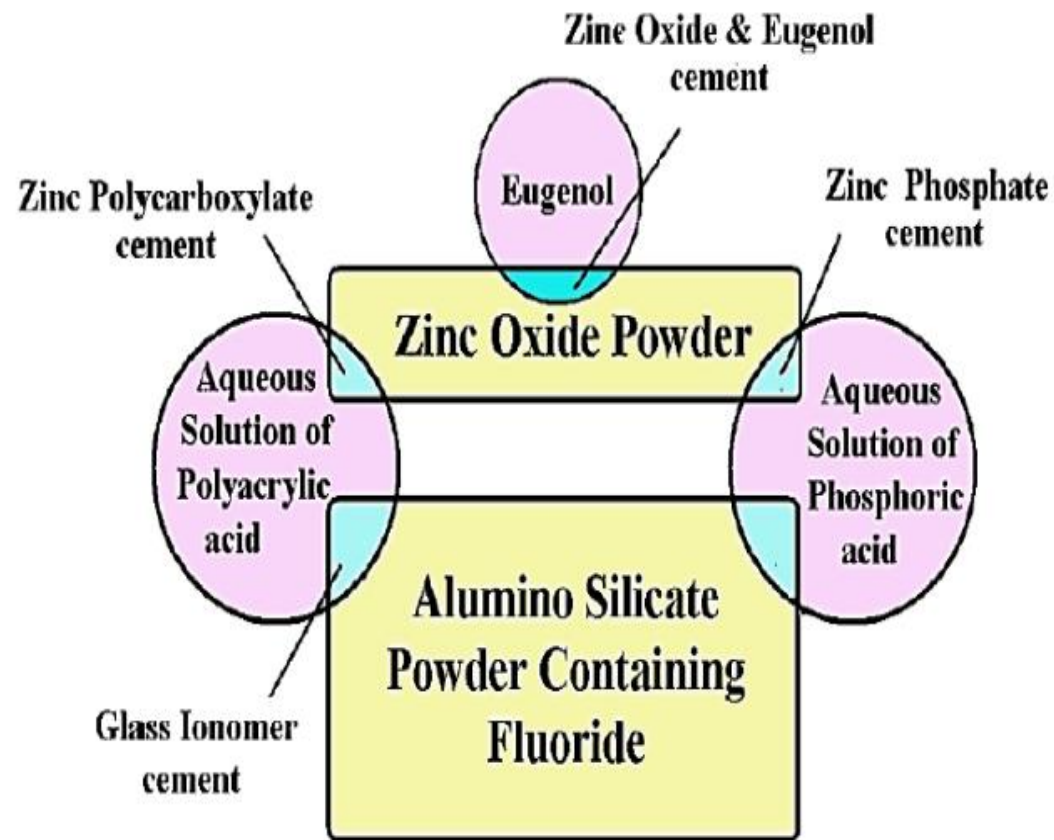


Figure 2. The main ingredients in the composition of the oil-based and water-based cements.

Zinc Oxide and Eugenol Cement

► It is one of the earliest cements used in dentistry.

Mode of supply:

- A. Powder and liquid: for cementation.
- B. Ready-made single paste: as temporary filling.
- C. Two paste: for temporary cementation.



Zinc Oxide and Eugenol Cement

Composition of Type I :

A. Powder:

Zinc oxide + 8 % zinc as accelerator+ Rosin to reduce brittleness and increase working time and strength.

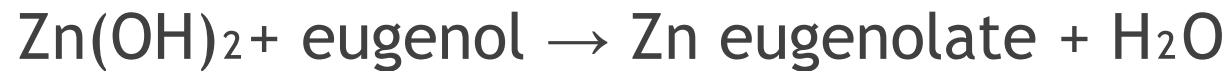
B. Liquid:

Eugenol + acetic acid as accelerator



Zinc Oxide and Eugenol Cement

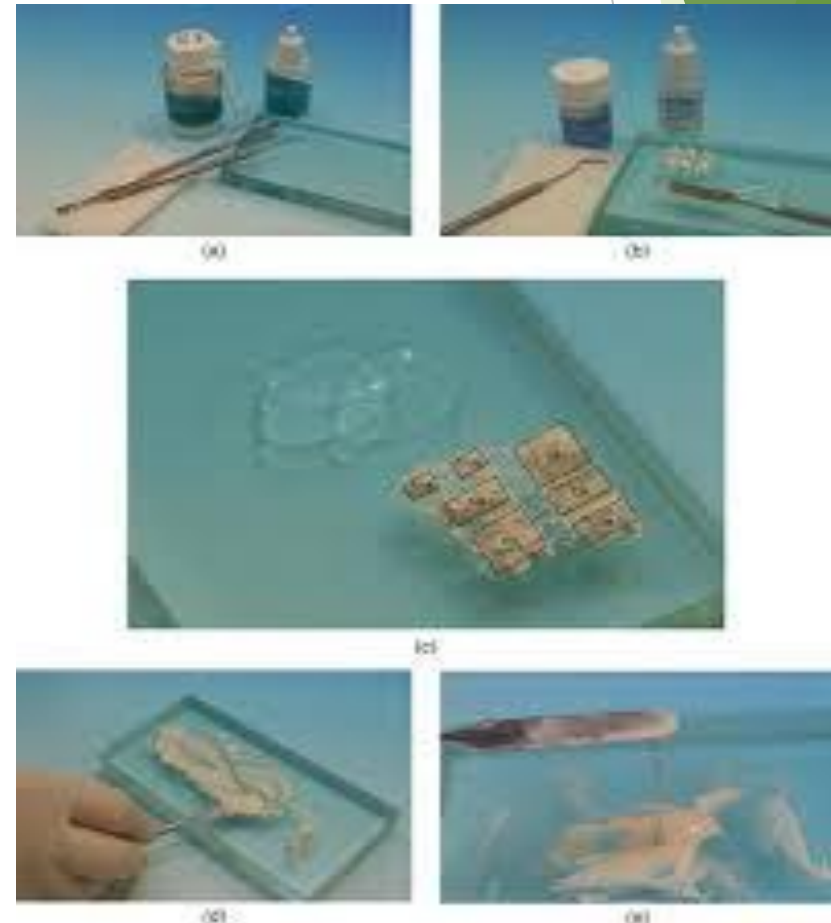
Setting reaction:



It is called **chelation** reaction.

Microstructure:

The set material consists of unreacted zinc oxide and free eugenol in a zinc eugenolate matrix.



Zinc Oxide and Eugenol Cement

Setting reaction:

- ▶ Water is essential for the setting reaction to occurs as ZnO cannot react with eugenol. Only zinc hydroxide can react with eugenol.
- ▶ Humidity and heat accelerate the reaction.
- ▶ Ready -made single paste contains no water and start to set after contact with moisture inside patient mouth.



Zinc Oxide and Eugenol Cement

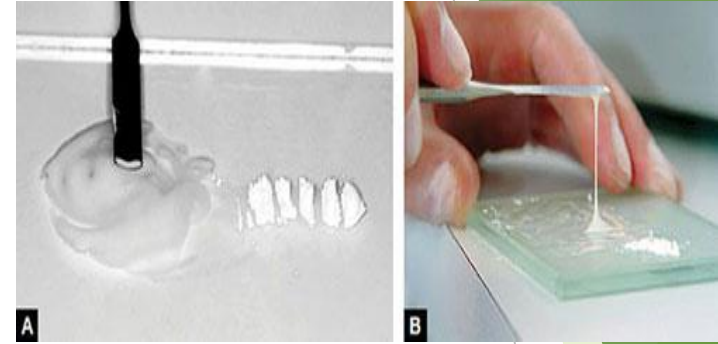
Manipulation:

Powder and liquid:

- ▶ Mixing using stainless-steel spatula over glass slab or oil resistant pad.
- ▶ The P/L ratio depends on the required viscosity and use.
- ▶ Powder is added incrementally to the liquid.

Two pastes:

- ▶ Dispense equal lengths and mix until obtain homogenous color.



Zinc Oxide and Eugenol Cement

Properties:

1. Biological properties:

- ▶ The set cement is **neutral** (pH=7).
- ▶ The eugenol has sedative , palliative and obtundent effect on the pulp.
- ▶ Eugenol has a **bacteriostatic** action.
- ▶ Avoid high conc. Of eugenol to avoid inflammation.

Zinc Oxide and Eugenol Cement

Properties:

2. Consistency & film thickness:

- ▶ Film thickness= 40 microns which is high compared to the other cements.

3. Solubility:

- ▶ **High solubility** due to leaching out of unreacted eugenol and its replacing by water. The water disintegrate the matrix.

Zinc Oxide and Eugenol Cement

Properties:

4. Strength:

- ▶ Poor mechanical properties
- ▶ Compressive strength = 15 MPa
- ▶ Tensile strength = 5 Mpa

5. Bonding:

- ▶ Mechanical interlocking.

6. Optical properties :

- ▶ Opaque due to the presence of unreacted ZnO particles in the set cement.

Zinc Oxide and Eugenol Cement

Modifications:

1. *Polymer reinforced cement Type II:*

Zinc oxide and eugenol cement is reinforced by addition of:

- a) 20% poly-methyl-methacrylate to the powder.
- b) 10% polystyrene to the liquid.

Compressive strength = 38MPa

Can be used as :

- 1. Final cement
- 2. Cement base
- 3. Temporary filling

Zinc Oxide and Eugenol Cement

Modifications:

2. *EBA cement Type II* :

It is modified by addition of:

- a) 30 % alumina to the powder → increase strength.
- b) 6% rosin and copolymers to the powder → increase plasticity.
- c) 62.5 % ethoxy benzoic acid (EBA) to liquid promotes formation of stronger , crystalline matrix.

Compressive strength= 85 Mpa.

Can be used as :

- 1. Base under amalgam
- 2. Final cementation
- 3. Temporary filling



Zinc Oxide and Eugenol Cement

Modifications:

3. Rapid set zinc oxide eugenol cement:

By:

- Modifying powder particle size & shape
- P/W ratio
- Increasing amount of accelerators



4. Eugenol free cement:

- This form is based on oils other than eugenol.
- Suitable for patients sensitive to eugenol

Zinc Oxide and Eugenol Cement

Limitations:

Zinc oxide eugenol cement:

- 1-Shouldn't be used as base under composite resin **why???**
- 2-Not as temporary filling for teeth that will be restored with composite resin. **Why???**
- 3- Not as temporary cement for crown on teeth to be restored using composite resin cement.

Why??? Because : ZnO is opaque

Eugenol inhibit polymerization reaction.

Zinc Oxide and Eugenol Cement

Limitations:

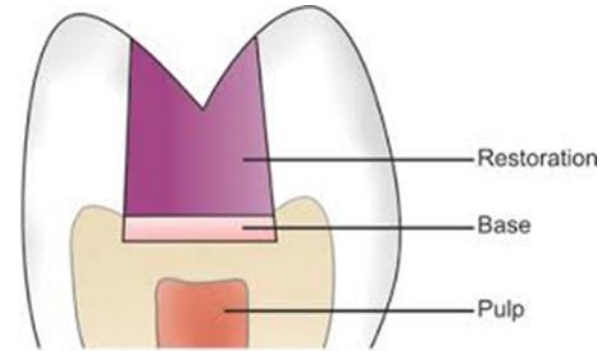
- ▶ The unmodified form has lower mechanical properties and high solubility. Therefore, it cannot be used base under amalgam restoration **why???**
- ▶ Eugenol containing cement is contra-indicated to be used for temporary cementation for teeth to be restored with all ceramic restoration , **why?????**

As all ceramic restoration is cemented finally by resin cement, any remnants from eugenol will inhibit the polymerization reaction of final resin cement .

Zinc Oxide and Eugenol Cement

Uses:

- ▶ Temporary cement “TYPE I”.
- ▶ Permanent cement (the modified types TYPE II).
- ▶ Base under restoration.
- ▶ Temporary filling material “TYPE I”.
- ▶ Surgical & periodontal pack.
- ▶ Root canal sealers.
- ▶ Inelastic impression material.



Water- based cements

1. Zinc Phosphate cement
2. Zinc Polycarboxylate cement
3. Glass-ionomer cement

Zinc Phosphate Cement

- It is one of the oldest cements that is available nowadays. This is because its long history of clinical success and good handling properties.

Mode of supply:

Powder and liquid



Zinc Phosphate Cement

Composition:

1. Powder:

- a) Zinc oxide(main ingredient).
- b) Magnesium oxide (gives a white color to the powder and control the setting reaction).
- c) Silica and alumina (improve mechanical properties)

2. liquid:

- ▶ Aqueous solution of phosphoric acid buffered by adding small quantities of zinc oxide or aluminum oxide.
- ▶ The oxides in the powder form a buffer with the liquid to control the working time.

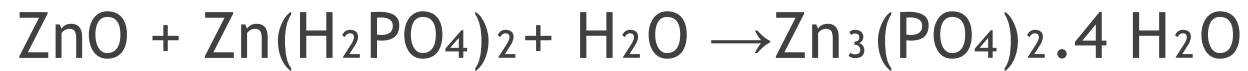
Zinc Phosphate Cement

Setting reaction:

- ▶ It is an **acid-base** reaction
- ▶ The acid dissolves the superficial layer of the particles of the powder and forms zinc phosphate.



- ▶ The second phase of the reaction occurs to produce hydrated zinc phosphate matrix.



- ▶ The reaction is **rapid** and **exothermic**

Zinc Phosphate Cement

Manipulation:

Powder/Liquid depends on the using purpose.

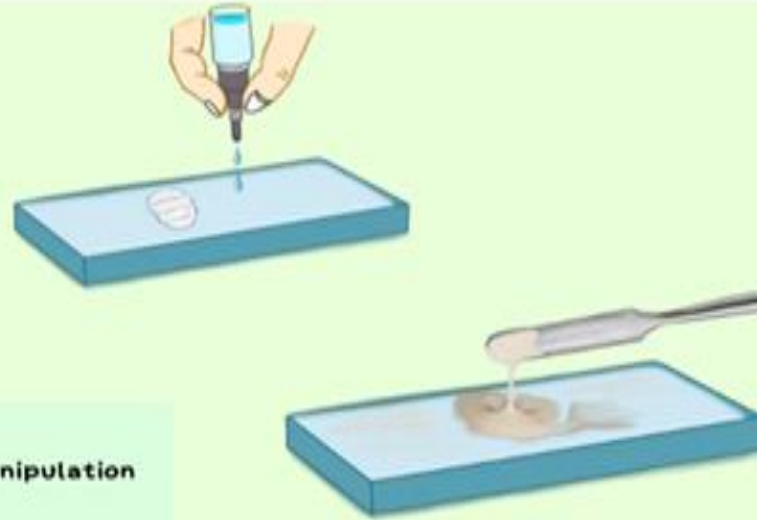
- ▶ Thin consistency for luting.
- ▶ Thick consistency for base.

Mixing is done using stainless steel spatula and over glass slab.

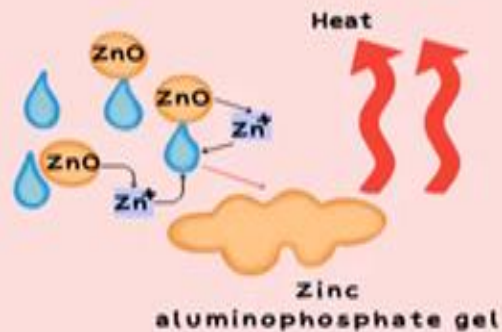
Zinc Phosphate Cement



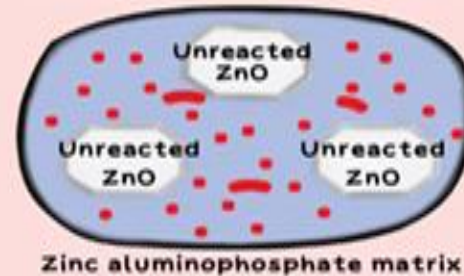
Zinc phosphate cement



Manipulation



Zinc phosphate cement - set material



Setting reaction and set material

Zinc Phosphate Cement

Manipulation:

The liquid is kept in tightly closed bottle **why???** to avoid loss and gain of water

Loss of water from the bottle by evaporation will cause the loss of the liquid chemical equilibrium.



precipitation of zinc and aluminum ions in the form of zinc and aluminum phosphate crystals resulting in liquid clouding.



the liquid will become unstable for mixing and delay the setting reaction of the cement.



Zinc Phosphate Cement

Manipulation:

- ▶ On the other hand, gaining a little amount of water by the liquid should also be avoided



it will accelerate the rate of ionization



decrease the setting time of the cement.

Zinc Phosphate Cement

Manipulation:

- ▶ To avoid loss of water from the liquid:
 - a) Keep the bottle tightly closed.
 - b) Dispense the cement just before use.
 - c) Dispense the powder followed by the liquid.

- ▶ The setting reaction is an exothermic reaction.
- ▶ The liberated heat accelerates the setting reaction which results in decreasing the working time.
- ▶ Therefore, heat should be dissipated.

Zinc Phosphate Cement

Manipulation:

- ▶ The heat should be dissipated by:
 - a) Mix over a wide area.
 - b) Mix incrementally.
 - c) Use a thick glass slab.
 - d) Use a cool glass slab.
- ▶ The temperature of the cooled glass slab should be above the **dew point** to prevent water condensation from the atmosphere  Water will be incorporated into the mix and accelerate setting time.
- ▶ This excess water alters the cement properties.

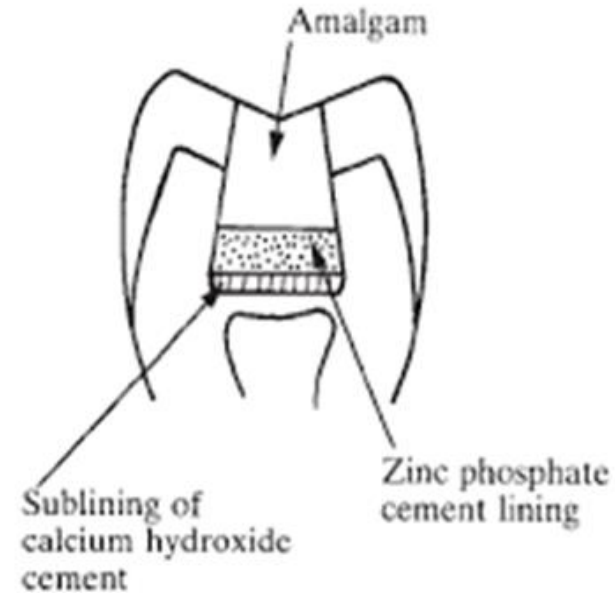


Zinc Phosphate Cement

Properties:

1. *Biological properties:*

- ▶ It is highly acidic
- ▶ The PH after mixing is 1.3 -3.6
- ▶ After 1 hour it rises to 6
- ▶ Then become neutral after 48 hours.
- ▶ If used as a base in deep cavities, it should be preceded by a sub-base (calcium hydroxide).
- ▶ If it is used for cementation, the patient may complain from pain for few minutes.



Zinc Phosphate Cement

Properties:

2. *Consistency and film thickness:*

- ▶ Thin film thickness = 25 microns

3. *Solubility:*

- ▶ Less soluble than ZnO eugenol.

4. *Mechanical properties:*

- ▶ It is **brittle material** (compressive strength = 95-135 MPa , Tensile strength = 3-5 MPa).
- ▶ Reaches about half of its strength after 10 minutes.
- ▶ Reaches final strength values after 24h.
- ▶ Has highest modulus of elasticity similar to dentine.

Zinc Phosphate Cement

Properties:

5. bonding:

- ▶ Mechanical interlocking.

6. Optical properties

- ▶ Opaque due to the presence of unreacted zinc oxide particles.

7. Thermal and electrical conductivity:

- ▶ Good thermal and electrical insulator but not effective as dentine.

Zinc Phosphate Cement

Uses:

1. **Permanent cement** for **metallic and metal-ceramic** restorations. **Why ??**
2. **Base** under amalgam. **Why???** The highest modulus of elasticity among all cements which is similar to dentin.
3. Temporary filling material.

Zinc Polycarboxylate

- ▶ It is the first cement that produce chemical bond with tooth structure.

Mode of supply:

1. Powder and liquid
2. Reproportioned capsules
3. Water settable cement



Composition :

1. *Powder:*
 - ▶ Like zinc phosphate cement (Zinc oxide+ magnesium oxide +silica + alumina + stannous fluoride)

Zinc Polycarboxylate

Composition:

2.Liquid :

- ▶ 30% - 40 % aqueous solution of copolymer of polacrylic acid and other unsaturated carboxylic acids (itaconic and maleic acids) to stabilize the liquid from gelation during storage.
- ▶ Sodium hydroxide to adjust the PH and viscosity.
- ▶ Tartaric acid to extend the working time.

Zinc Polycarboxylate

Composition:

3. *Water settable cement:*

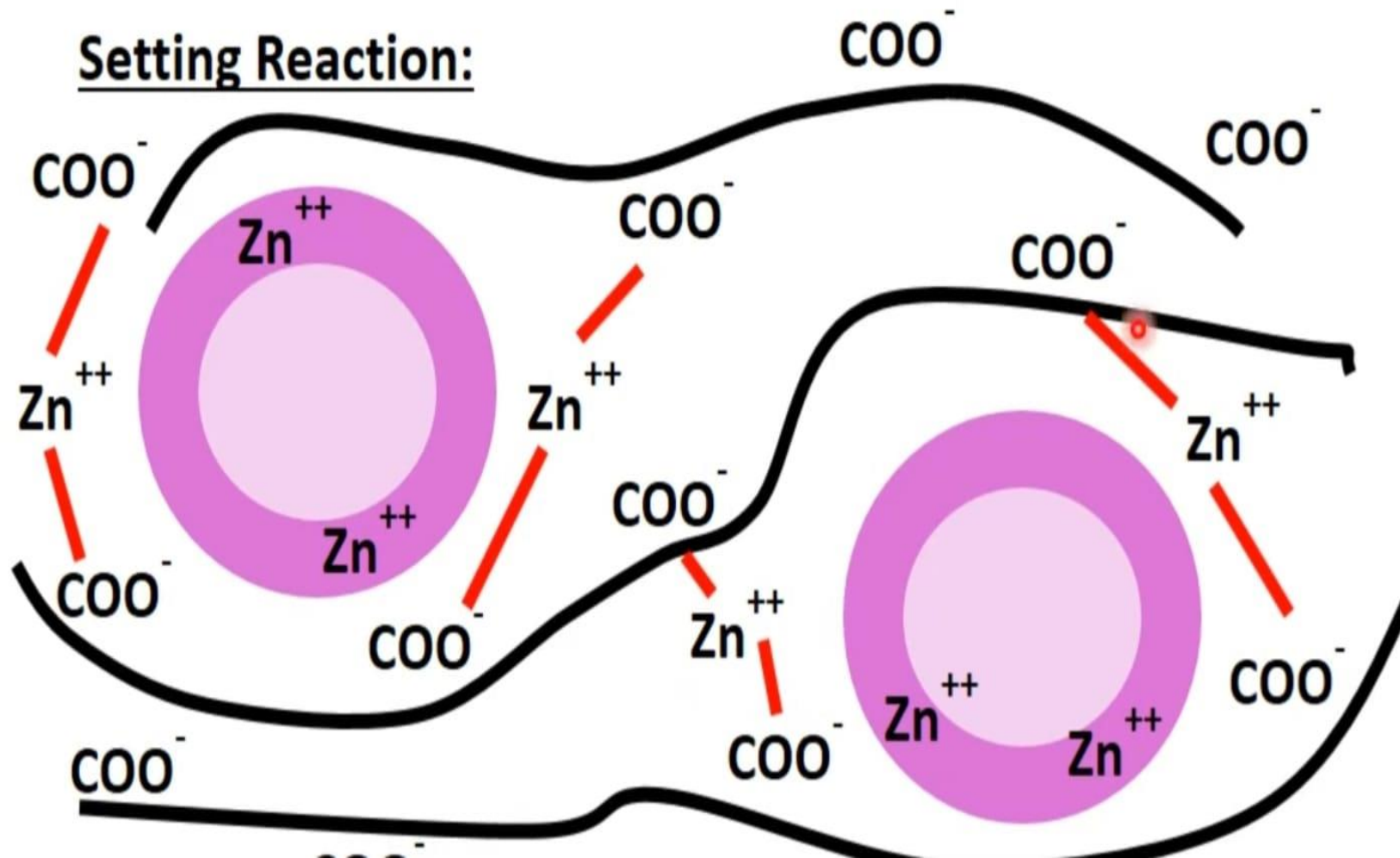
- ▶ The polycarboxylic acid is freeze- dried and then added to the powder.
- ▶ The liquid of this formula is distilled water or diluted solution of tartaric acid.
- ▶ This formula has an extended working time compared to traditional form.

Zinc Polycarboxylate

Setting reaction:

- ▶ It is an **acid-base** reaction.
- ▶ The ionized copolymer attack the superficial layer of zinc oxide powder and release zinc ions.
- ▶ Zinc ions cross- link the copolymer chains forming a matrix of zinc polyacrylate (zinc polycarboxylate).
- ▶ It is **faster** than setting reaction of zinc phosphate cement.

Zinc Polycarboxylate



Zinc Polycarboxylate

Manipulation:

- ▶ P/L depends on the using purpose.
- ▶ Mixing is done using plastic spatula **Why???** and over glass slab or non - absorbent paper .
- ▶ Stainless steel spatula can react chemically with the cement.
- ▶ Like zinc phosphate cement, the cement should be dispensed just before mixing. **Why??** To prevent water evaporation and subsequent thickening.

Zinc Polycarboxylate

Manipulation:

- ▶ The working time of the cement is **shorter** than that of zinc phosphate cement. The working time can be extended by:
 1. Incorporate half of the powder to liquid at a time.
 2. Refrigerate the powder.
 3. The recent formulations have longer working time as the manufacturers optimized the amount of tartaric acid.

Zinc Polycarboxylate

Manipulation:

- ▶ Cooling the liquid or using a cool glass slab increases its viscosity, so it is not recommended.
- ▶ The cement should be used while it still has its glossy surface **why???** as the loss of its surface gloss indicate initial setting of the cement and absence of free carboxylic group that are essential for chemical bond.

Zinc Polycarboxylate

Properties:

1. Biological properties:

Although its initial low PH, it is more biocompatible than zinc phosphate cement due to: why???

- a) Polyacrylic acid is weaker acid compared to phosphoric acid.
- b) The rapid rise of the PH to neutrality.
- c) The large molecular size of polyacrylic acid prevent its penetration into dentinal tubules.
- The chemical bond with the tooth structure provides a barrier against ingress of the bacteria.

Zinc Polycarboxylate

2. *Consistency and film thickness:*

- ▶ Thin film thickness = 25-45 microns > Zn phosphate

3. *Solubility:*

- ▶ Low solubility 0.01% to 0.02% after 24 hours < Zn phosphate.

4. *Mechanical properties:*

- ▶ Compressive strength < Zn phosphate cement (55-95 Mpa).
- ▶ Tensile strength slightly > Zn phosphate cement (3-7 Mpa).

Zinc Polycarboxylate

5. *bonding:*

- ▶ It bonds **chemically** with enamel and dentine.
- ▶ The carboxylic group bonds with the calcium of the tooth structure.
- ▶ The bond to enamel is higher than that to dentine due to high mineral content.
- ▶ Chemical bonding with base metal alloy.
- ▶ Mechanical bonding with gold alloys as gold is inert so sandblasting is done to increase surface roughness and mechanical interlocking.

6. *Optical properties:*

- ▶ **Opaque** due to the presence of unreacted zinc oxide particles.

Zinc Polycarboxylate

Uses:

- ▶ Permanent cement for metallic and metal-ceramic restorations.
- ▶ Base under amalgam

N.B.

Zinc polycarboxylate cement should not be used for temporary cementation **why??** because it bonds chemically to tooth structure.

Classification of Dental Cements

1. According to bonding with tooth structure:

Micromechanical interlocking	Chemical bonding
• Zinc oxide and eugenol cement • Zinc phosphate cement • Conventional resin cement • Calcium hydroxide liner	• Zinc polycarboxylate cement • Glass Ionomer cement • Adhesive resin cement

Classification of Dental Cements

2. According to setting reaction:

Acid - base reaction	Polymerization reaction	Acid base & polymerization reaction
• Zinc oxide & eugenol cement • Zinc phosphate cement • Zinc polycarboxylate cement • Glass Ionomer cement	• Resin cement	• Resin modified Glass Ionomer cement

Classification of Dental Cements

3. According to chemistry of the powder (Historical classification):

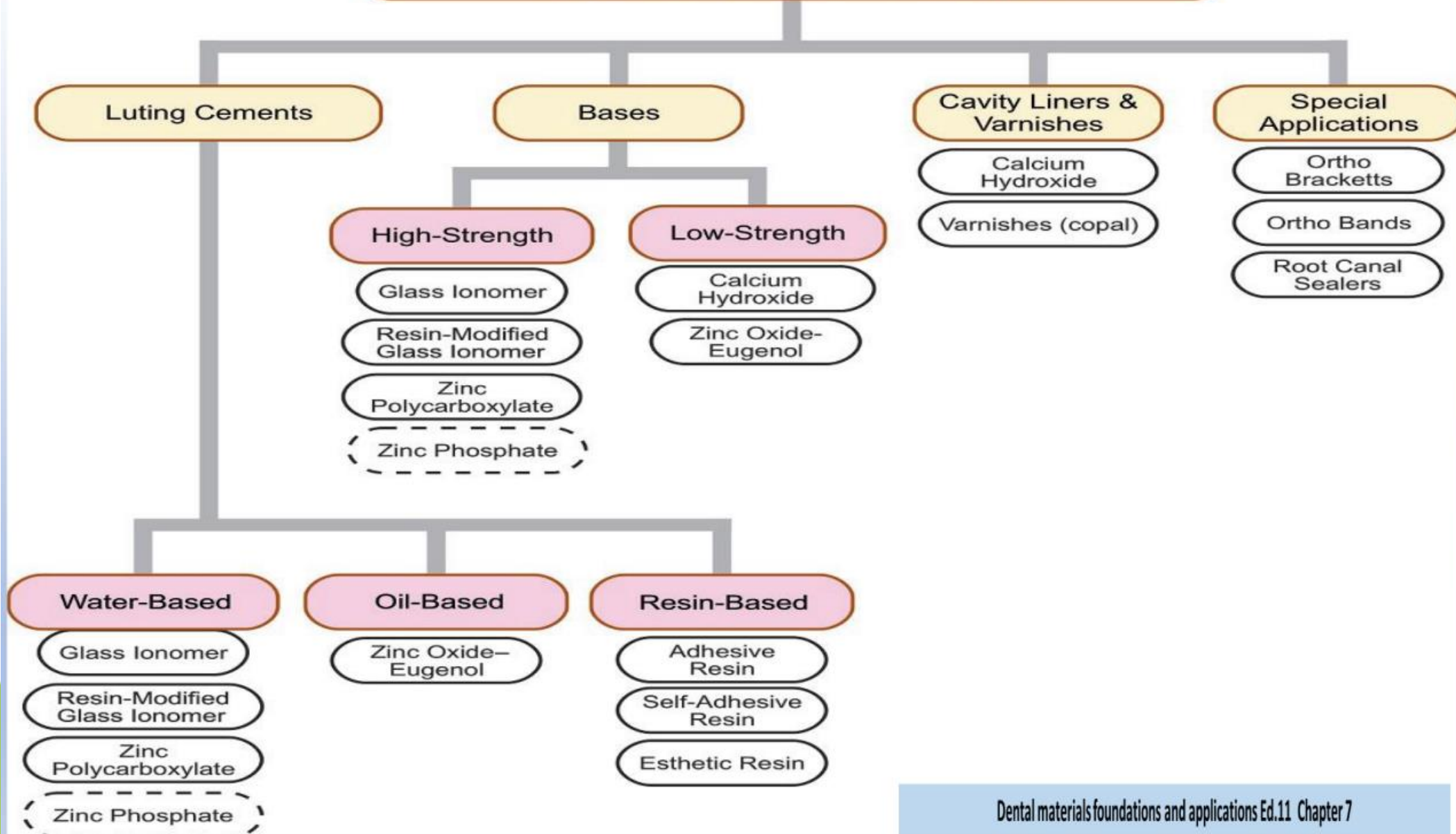
Cements based on zinc oxide powder	Cements based on Fluoro - alumino - silicate glass powder
• Zinc oxide and eugenol cement • Zinc phosphate cement • Zinc polycarboxylate cement	• Glass ionomer cement • Silicate cement

Classification of Dental Cements

4. According to their chief chemical ingredients:

Oil-based cement	Water-based cement	Resin-based cement
• Zinc oxide & eugenol cement	• Zinc phosphate cement • Zinc polycarboxylate cement • Glass ionomer cement	• Resin cement

Types of Cements in Dentistry



Thank You